

VISHVAKARMA.OS

v5.7 [QUANTUM-SUSHUPTI EDITION]

वश्वकर्मा

Complete Technical Overview & Architecture Reference
The World's First Bio-Adaptive, Vedic-Industrial Architecture OS

iPad-First | WebGPU | PWA | Real-Time Collaboration

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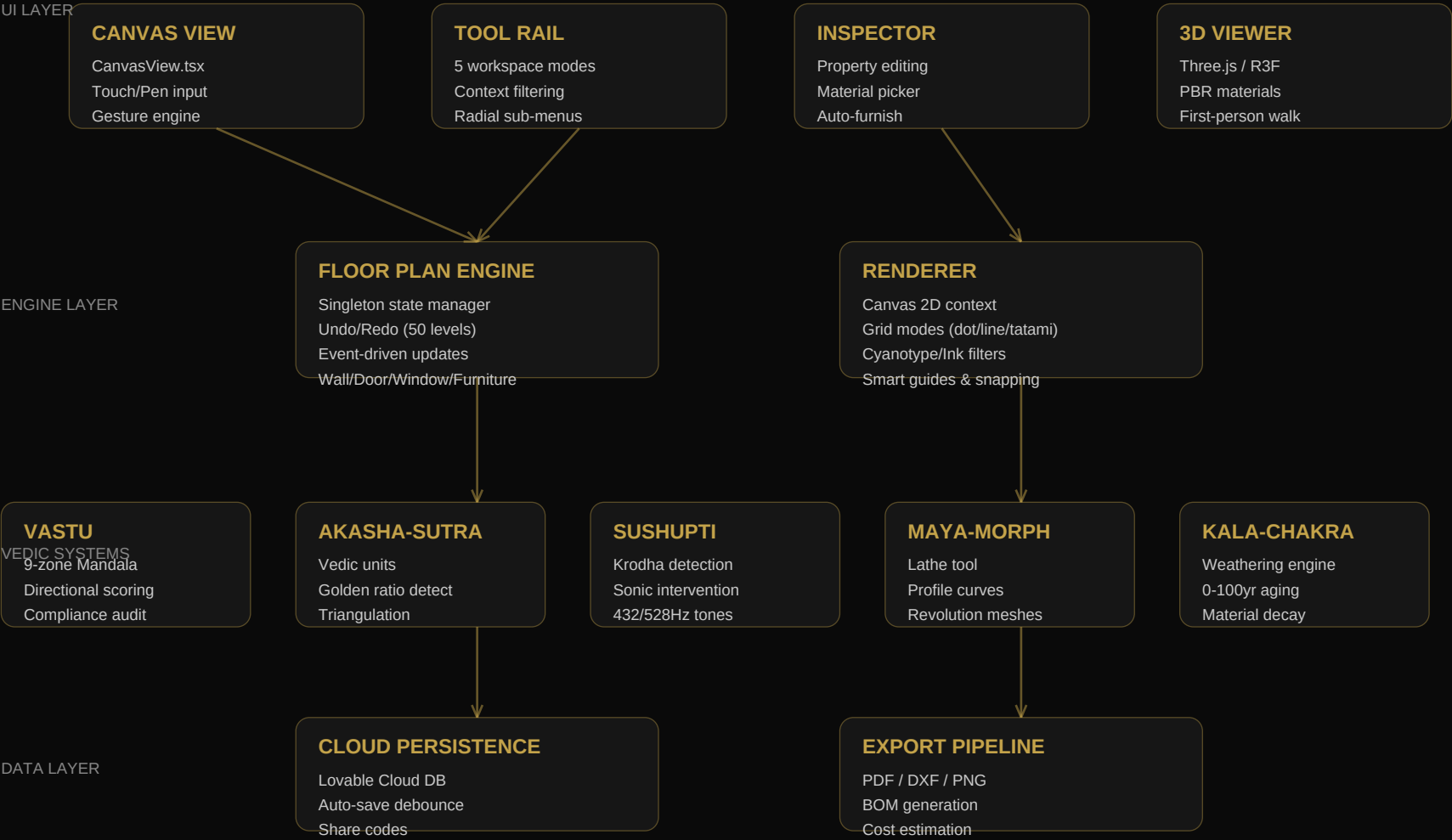
1. EXECUTIVE SUMMARY & WORLD FIRSTS

VISHVAKARMA.OS is the world's first bio-adaptive, Vedic-industrial architecture operating system. Built as an iPad-first PWA, it combines precision 2D CAD drafting with real-time 3D PBR visualization, ancient Vastu Shastra spatial science, and a cognitive wellness engine (Sushupti) that monitors architect focus states. The system features 77+ integrated tools across 5 workspace modes.

WORLD FIRSTS

- **Kala-Chakra Temporal Weathering**
Age materials 0-100 years with procedural shaders in real-time 3D
- **Prana-Gati Airflow Simulation**
CFD-lite wind flow visualization overlaid on floor plans
- **Jala-Maya Water Caustics**
Real-time dancing light patterns from pools projected onto walls
- **Sushupti Bio-Adaptive Engine**
Detects architect frustration via cursor telemetry, triggers sonic reset
- **Akasha-Cast State Mirroring**
Cross-device real-time design broadcasting without screen-sharing
- **Shilpa-Bindu Construction Joints**
Procedural tenon-mortise/rivet details at 500%+ zoom
- **Vedic Unit System**
Angula/Hasta/Dhanush measurements with Devanagari labels
- **Sacred Sound Integration**
432Hz/528Hz Solfeggio tones wired to UI interactions

2. SYSTEM ARCHITECTURE DIAGRAM



3. CORE ENGINE & TYPE SYSTEM

FloorPlanEngine (Singleton)

- Central state manager for all 2D entities
- Manages: walls, doors, windows, furniture, rooms, measurements
- Electrical symbols, plumbing runs, staircases, niches, ceiling zones
- Undo/Redo stack with 50-level history
- Event system: onChange callbacks for reactive UI updates
- Auto-save via debounced cloud persistence

Type System (types.ts)

- Point {x, y} - Base coordinate type
- Wall {id, start, end, thickness, color, loadBearing, material}
- Door/Window {id, wallId, offset, width, swing/type}
- FurnitureItem {id, type, x, y, w, h, rotation, material, roomId}
- Room {id, label, type, polygon, floorMaterial}
- EditorState {tool, mode, showGrid, darkMode, gridMode, showVastu...}

Geometry Engine (geometry.ts)

- distance(), wallLength(), wallDirection(), wallNormal()
- snapToGrid(), snapAngle() (15-degree increments)
- snapEndpoint() - snap to existing wall endpoints
- polygonArea() - Shoelace formula for room areas
- isGoldenRatio() - PHI detection within 4% tolerance
- pointInPolygon() - Ray-casting hit test

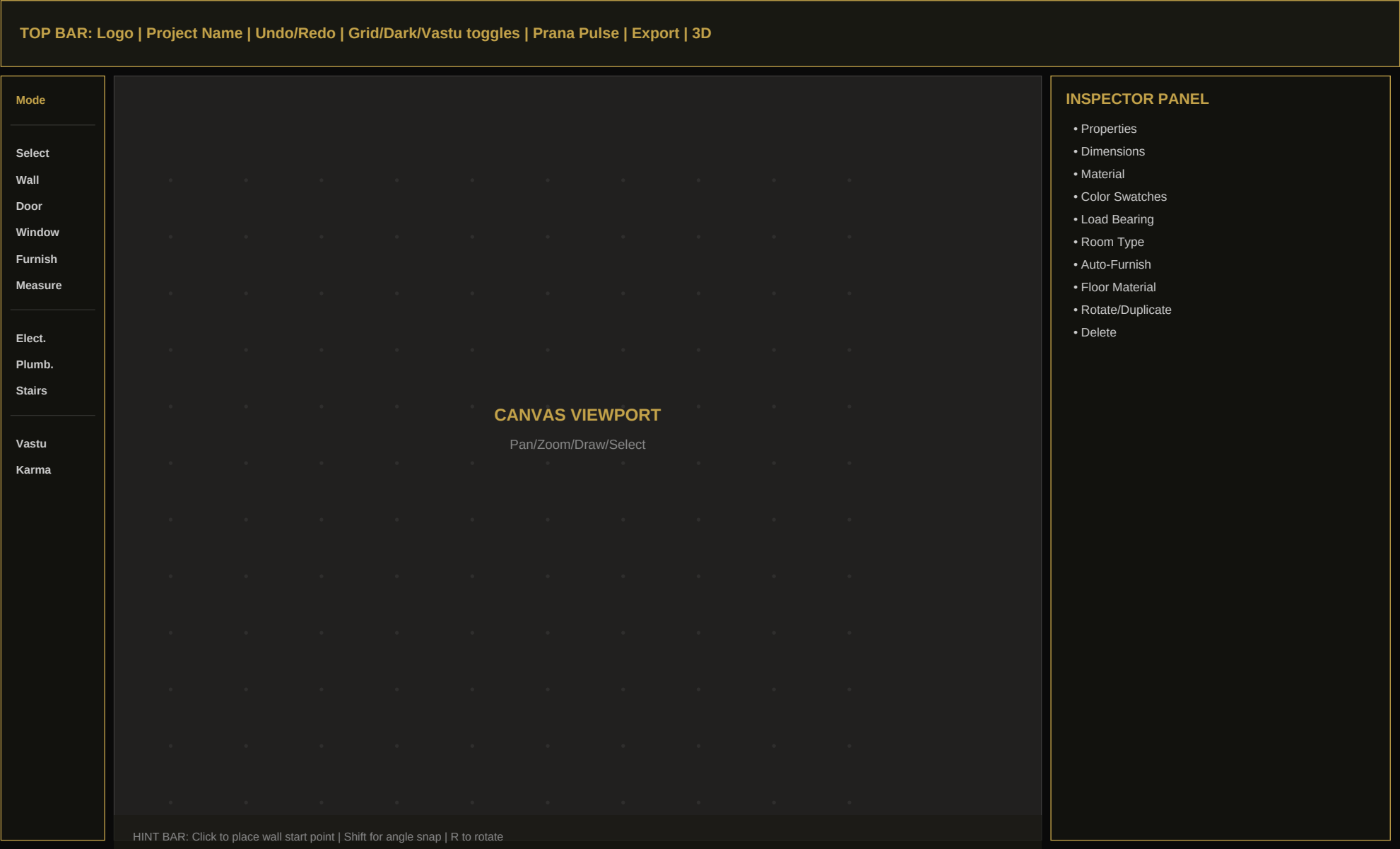
4. WORKSPACE MODES & UI FLOW

Mode	Tools Available	Top Bar Stats	3D Behavior
DRAFTING	Walls, Doors, Windows, Measure, Staircase	Floor Area / Perimeter	Flat Grid (Z=0)
MEP	Electrical, Plumbing, HVAC symbols	Wire/Pipe Length totals	Overlay on plan
INTERIOR	Furniture, Lighting, PBR materials	Cost / Lux levels	Orthographic 3D
LANDSCAPE	Terrain sculpt, Paths, Foliage	Cut & Fill / Slope %	Global terrain mesh
AVATAR	First-person walk, Measure, Swap	Eye-level compass	First-person locked

USER FLOW

1. Auth Portal: "Anahata" gateway with animated Sri Yantra + sacred sound (528Hz)
2. "HariOm" Creation Ritual: 3-phase project creation with compass calibration
3. Canvas Workspace: Obsidian drafting surface with gold-etched grid overlay
4. Tool Selection: Mode-filtered rail on left, context-aware Inspector on right
5. Drawing: Snap-to-grid walls with real-time Vedic unit display
6. Room Detection: Automatic closed-polygon room creation with type assignment
7. Auto-Furnish: One-click room furnishing based on room type & dimensions
8. 3D Preview: Toggle to PBR 3D view with material painting and first-person walk
9. Analysis: Vastu compliance, acoustic heatmap, airflow simulation overlays
10. Export: PDF blueprints, DXF CAD files, PNG renders, BOM cost estimates

5. UI ARCHITECTURE DIAGRAM



6. 2D DRAFTING ENGINE

Wall System

- Click-to-click wall drawing with real-time preview
- Configurable thickness (4-40px) and color
- Load-bearing designation with visual indicator
- Auto-snap to grid (20px) and existing endpoints
- Angle snapping at 15-degree increments (0/15/30/45/60/75/90)
- Material assignment (concrete, brick, stone, glass, wood, steel)

Doors & Windows

- Place on any wall segment with offset positioning
- Door: configurable swing direction (left/right), width 40-120px
- Window: adjustable width 30-100px
- Arc swing visualization for doors

Room Detection & Management

- Automatic closed-polygon detection from wall intersections
- Room types: bedroom, bathroom, kitchen, living, dining, study, storage, hallway, balcony, garage
- Area calculation via Shoelace formula with m^2 display
- Floor material assignment (marble, teak, concrete, tile, carpet, bamboo, terrazzo)
- Golden ratio detection on room bounding boxes

Furniture System

- 10 types: bed, sofa, table, chair, wardrobe, bathtub, toilet, kitchen, desk, dining
- Rotation: 0/90/180/270 degrees with R key
- Duplication with Cmd+D
- Auto-furnish: one-click room population based on type and dimensions
- Drag-to-reposition with collision awareness

Measurement & Annotation

- Point-to-point dimension lines with formatted distance display
- Akasha-Sutra: Vedic unit overlay (Angula/Hasta/Dhanush)
- Golden ratio relationship detection between walls
- Parallel/perpendicular alignment analysis

7. MEP SYSTEMS

Electrical System

- Symbol types: outlet, switch, light, fan, panel, data, smoke
- Wire run calculation with shortest-path routing to panels
- Cost estimation per meter of wire
- Visual symbols rendered at wall positions

Plumbing System

- Pipe routing along walls with connection point specification
- Manning equation flow calculation: $V = (1/n) * R^{(2/3)} * S^{(1/2)}$
- Material roughness coefficients: PVC (0.009), Copper (0.011), Cast Iron (0.013)
- Bend counting and total pipe length calculation
- Cost estimation for pipes, fittings, and labor

Nesting Optimizer

- Strip-packing algorithm for wall panel material optimization
- Configurable sheet dimensions (standard 2440x1220mm)
- Waste percentage calculation and visual layout
- Multi-sheet distribution with piece identification

Routing Panel

- Plumbing route: total length, segments, bends summary
- Electrical: shortest wire runs from symbols to panels
- Combined material cost estimation

8. 3D VIEWER & PBR MATERIALS

Rendering Stack

- Three.js + React Three Fiber (@react-three/fiber)
- Drei helpers for controls, environment, lighting
- OrbitControls with configurable min/max distance
- Environment map (apartment preset) for realistic reflections
- SSAO, Bloom post-processing for cinematic quality

PBR Material Library (16 materials)

- Stone: Carrara Marble, Indian Sandstone, Kota Stone
- Wood: Teak, Rosewood (Sheesham), Bamboo
- Metal: Brass, Wrought Iron, Copper
- Modern: Exposed Concrete, Terracotta, Glass
- Specialty: Jali Lattice, Corten Steel, Rammed Earth, Lime Plaster
- Each: procedural color/normal/roughness maps, metalness, pattern generation

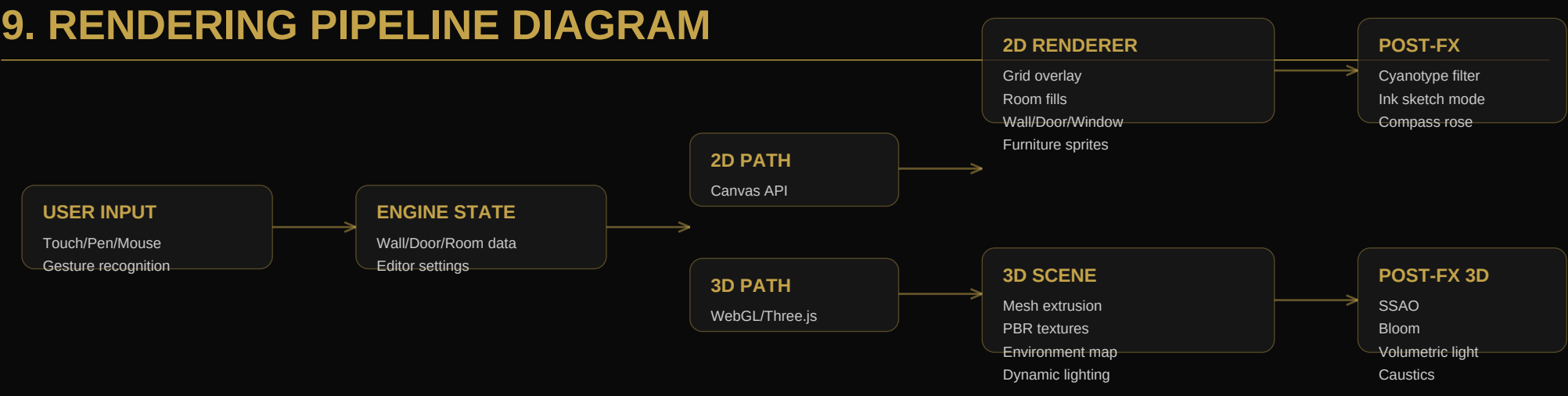
3D Features

- Wall extrusion from 2D plan with configurable height
- Door/window cutouts with frame geometry
- Floor plane with material tiling
- Furniture placement matching 2D positions
- Multi-floor system with floor switching
- Section box for architectural cutaway views
- Paint mode: click surfaces to apply PBR materials

First-Person Walker

- Eye-level camera at 1.6m height
- WASD + mouse look (pointer lock)
- Virtual joystick for touch devices
- Wall collision with sliding response
- Collision vignette visual feedback

9. RENDERING PIPELINE DIAGRAM



OVERLAY SYSTEMS



10. VEDIC SYSTEMS

Vastu Shastra Compliance Engine

- 9-zone Padmasana Mandala (3x3 grid aligned to compass)
- Directional scoring: NE (water/Jala), SE (fire/Agni), SW (earth/Prithvi), NW (air/Vayu)
- Brahmasthana (center): must remain open for optimal Prana flow
- Room-type validation against Vastu zone recommendations
- Overall compliance score with per-zone breakdown
- Visual overlay with directional colors and Sanskrit zone labels

Akasha-Sutra Measurement System

- Vedic unit conversion: 1 Angula = 1.9cm, 1 Hasta = 45cm, 1 Dhanush = 180cm
- Devanagari labels: Angula (अंगुल), Hasta
- Golden Ratio (PHI = 1.618) detection between wall lengths
- Quantum snapping: parallel, perpendicular, equal-length alignment detection
- LiDAR True Distance: 3D-aware measurement with curvature correction

Panchatattva (Five Elements) Panel

- Maps room elements to: Prithvi (Earth), Jala (Water), Agni (Fire), Vayu (Air), Akasha (Space)
- Element balance scoring for each room
- Remediation suggestions for elemental imbalance

Karma Timeline

- Circular mandala-style version control visualization
- Every action recorded as a "Karmic Seed"
- Branch navigation for exploring alternative designs

11. SUSHUPTI BIO-ADAPTIVE ENGINE

Krodha (Frustration) Detection

- Mouse Thrash: 8+ direction changes within 1.5 seconds
- Rage Click: 5+ high-velocity clicks within 2 seconds
- Undo Loop: 3+ undo/redo pairs on same area in 10 seconds
- Real-time telemetry via pointer move and click event analysis
- Configurable thresholds with exponential cooldown

Svara (Sound) Intervention Matrix

- 528Hz (Transformation): Plays on structural errors to aid problem-solving
- 432Hz (Universal Harmony): Triggered during mouse thrashes to lower heart rate
- 741Hz (Problem Solving): Activates during complex MEP/Vastu routing
- Binaural Alpha waves (8-13Hz): Background during Draft mode for deep focus
- Binaural Theta waves (4-8Hz): During 3D walkthroughs for spatial intuition

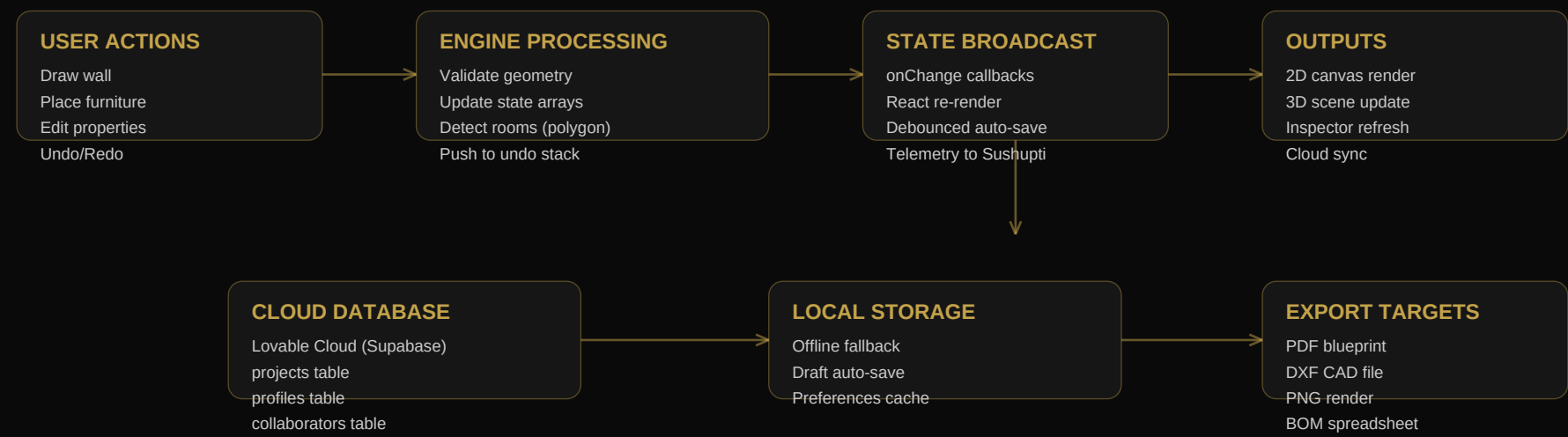
Prana Soundscapes

- Himalayan Stream: Flow speed matches drafting velocity
- Monsoon Rain on Jali: White-noise masking for office environments
- Temple Gongs: Milestone completion dopamine reset

Prana Pulse HUD

- Lotus icon in TopBar with color-coded state indication
- Glowing Gold: Flow state - optimal creative zone
- Pulsing Amber: Frustration detected - UI dims non-essentials
- Deep Obsidian: Zen mode - minimal sound, maximum focus
- "Breathe" overlay: Glassmorphic Shunya prompt on severe rage clicks
- 3-second breathing cycle with animated expand/contract circle

12. DATA FLOW DIAGRAM



DATABASE SCHEMA

projects	id, name, owner_id, data (JSON), is_public, share_code, created_at, updated_at
profiles	id, user_id, display_name, firm_name, avatar_url, created_at, updated_at
collaborators	id, project_id, user_id, role, created_at

RLS policies enforce owner/collaborator access. Security definer functions prevent recursive checks.

13. ADVANCED FEATURES MATRIX

Feature	Capability	Implementation
Kala-Chakra Weathering	0-100yr aging slider	Procedural Vibhuti shader
Prana-Gati Airflow	CFD-lite wind simulation	Particle streamlines on 2D
Jala-Maya Caustics	Water light refraction	Dancing patterns in 3D
Acoustic Heatmap	Sound reflection model	dB-colored overlay
Daylight Shadow	Sun position calculation	Shadow projection on plan
Shilpa-Bindu Joints	500%+ zoom detail	Tenon-mortise/rivet procedural
Maya-Morph Lathe	2D profile revolution	Pillars, domes, vases
Niche/Recess Tool	Boolean wall subtraction	LED coves, window seats
Ceiling Zone Lab	Tray/coffered geometry	False ceiling design
Sacred Sound	432/528Hz frequencies	UI interaction feedback
Auto-Furnish	Room-type recipes	One-click population
Golden Ratio Detect	PHI = 1.618 analysis	Real-time wall relationships
Grid Modes	Dot/Line/Cross/Tatami	4 professional grid styles
Canvas Filters	Cyanotype/Ink Sketch	Architectural presentation
Command Palette	Cmd+K universal search	Tools, furniture, materials
Radial Menu	Long-press context menu	Quick tool sub-selection
Command Dial	Touch scrub wheel	Precision value adjustment

14. LANDSCAPE & EXTERIOR

Terrain System

- 3D terrain mesh generation from heightmap data
- Procedural grass/earth texturing with normal maps
- Cut & fill volume calculation for earthworks
- Slope percentage display and analysis

Garden & Paths

- Garden path placement with material selection
- Procedural cobblestone/gravel patterns
- Path width and curvature control

Foliage Library

- Tree placement with 3D mesh generation
- Species: Banyan, Ashoka, Neem, Palm, custom
- Root Protection Zone visualization
- Seasonal foliage color variation

Water Features (Jala-Maya)

- Pool volume carving with "Subtract" brush
- Cubic meter water volume calculation for BOM
- Dynamic caustic light projection onto adjacent surfaces
- Infinite edge pool geometry support

Exterior Structures

- Cantilevered carports with structural load calc
- Alfresco/outdoor kitchen with MEP routing points
- Boundary walls and fencing
- Driveway and parking layout tools

15. EXPORT & COLLABORATION

PDF Export

- A4 landscape with title block, scale bar, room schedule table
- Canvas-to-JPEG-to-PDF pipeline (zero external dependencies)
- Project metadata: name, date, wall/door/window/furniture counts
- Room schedule with area, type, and wall count per room

DXF Export (AutoCAD Compatible)

- Walls as LINE entities with thickness
- Doors as ARC entities with swing direction
- Room labels as TEXT entities
- Layer organization: Walls, Doors, Windows, Furniture, Dimensions

PNG Export

- High-resolution canvas rendering (2x DPR)
- Includes all visible layers and overlays
- Transparent or white background options

BOM / Cost Estimation

- Material quantity takeoff from all elements
- Unit cost database for common Indian construction materials
- Labor cost estimation based on regional rates
- Carbon footprint "Purity Score" calculation

Collaboration (Akasha-Cast)

- Real-time state mirroring across devices
- Share codes for project access
- Collaborator roles: owner, editor, viewer
- Cloud persistence via Lovable Cloud

16. KEYBOARD SHORTCUTS & GLOSSARY

KEYBOARD SHORTCUTS

R	Rotate selected furniture	Cmd/Ctrl + K	Command palette
G	Toggle grid visibility	Cmd/Ctrl + E	Export modal
Del / Backspace	Delete selected element	Shift (while drawing)	Angle snap
Escape	Deselect / Cancel	3-finger tap	Command palette (touch)
Cmd/Ctrl + D	Duplicate selected	Long press	Radial context menu
Cmd/Ctrl + Z	Undo	Pinch	Zoom canvas
Cmd/Ctrl + Shift + Z	Redo	Two-finger drag	Pan canvas

SANSKRIT GLOSSARY

Vishvakarma	Divine architect of the universe	Yantra	Sacred geometric diagram
Vastu Shastra	Ancient science of architecture & spatial harmony	Mandala	Circular sacred pattern
Prana	Life force / vital energy	Dharma	Righteous duty / cosmic law
Sushupti	Deep sleep / equilibrium state	Maya	Illusion / creative power
Krodha	Frustration / anger	Jala	Water element
Akasha	Ether / space element	Agni	Fire element
Karma	Action / cause-and-effect cycle	Vayu	Wind / air element
Shunya	Void / zero / minimalist balance	Prithvi	Earth element

17. FEATURE CHECKLIST & STATUS

- | | |
|---|-------------------------------------|
| ✓ Wall drawing with snap & angle constraints | ✓ Kala-Chakra weathering engine |
| ✓ Door/Window placement on walls | ✓ Prana-Gati airflow simulation |
| ✓ Room auto-detection & type assignment | ✓ Jala-Maya water caustics (3D) |
| ✓ Furniture library (10 types) with rotation | ✓ Acoustic heatmap overlay |
| ✓ Auto-furnish by room type | ✓ Sushupti bio-adaptive engine |
| ✓ Measurement tool with Vedic units | ✓ Sacred sound (432/528/741Hz) |
| ✓ Vastu 9-zone compliance overlay | ✓ Prana Pulse HUD |
| ✓ 3D PBR viewer with 16 materials | ✓ Shilpa-Bindu construction joints |
| ✓ First-person walker with collision | ✓ Maya-Morph lathe tool |
| ✓ Electrical symbol placement & wire routing | ✓ Niche/recess wall carving |
| ✓ Plumbing pipe routing with Manning equation | ✓ Ceiling zone designer |
| ✓ Nesting material optimizer | ✓ Terrain & landscape tools |
| ✓ PDF/DXF/PNG export pipeline | ✓ Akasha-Cast device mirroring |
| ✓ Cost estimator (BOM) | ✓ Karma timeline (version control) |
| ✓ Cloud save & collaboration | ✓ iPad/Apple Pencil optimization |
| ✓ Command palette (Cmd+K) | ✓ PWA offline support |
| ✓ Radial menu & Command dial | ✓ HariOm creation ritual |
| ✓ Dark mode & grid mode switcher | ✓ Layers panel with visibility/lock |
| ✓ Cyanotype & Ink Sketch canvas filters | |

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The Foundry is Complete. The Creation is Sovereign.

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iPad-First | WebGPU-Ready | Bio-Adaptive | Vedic-Industrial

77+ Features | 5 Workspace Modes | 16 PBR Materials | Real-Time Collaboration